

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY**First Semester of B. Pharm Examination****University Theory Examination January 2018****MA121 Basic Mathematics****Date: 01/01/2018, Monday****Time: 10:00 a.m. to 01:00 p.m.****Maximum Marks: 80****Instructions:**

1. There are three sections in this question paper.
2. SECTION -I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION-I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION-I is 30 minutes. Answers to SECTION-I must be written during the first 30 minutes of the examination.
3. Answers to SECTION-II and SECTION-III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat sketches wherever necessary.
6. Use of scientific non-programable calculator is allowed.

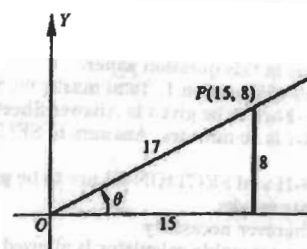
SECTION I**Q - 1 Choose the correct answer for the following questions each .****[20]**

1. Consider the sets $U = \{1, 2, \dots, 9\}$, $E = \{1, 2, 3, 4\}$, $F = \{2, 4, 6, 8\}$ and $G = \{3, 4, 5, 6\}$. Then $(E \cup F)' =$ _____
 (A) $\{3, 4\}$ (B) $\{2, 8\}$ (C) $\{5, 7, 9\}$ (D) ϕ
2. Suppose $n(U) = 110$, $n(E) = 30$, $n(F) = 35$ and $n(E \cap F) = 20$ then $n(E \cap F') =$ _____
 (A) 10 (B) 80 (C) 15 (D) 90
3. Let $E = \{1, 2\}$, and $F = \{a, b, c\}$ then $n(E \times F) =$ _____
 (A) 4 (B) 9 (C) 6 (D) 5
4. Consider the relations f, g and h on the set $D = \{1, 2, 3\}$ given by $f = \{(1, 3), (2, 3), (3, 1)\}$, $g = \{(1, 2), (3, 1)\}$ and $h = \{(1, 3), (2, 1), (1, 2), (3, 1)\}$ then,
 (A) f is not function from D into D . (C) g is function from D into D .
 (B) f is function from D into D . (D) h is function from D into D .
5. Let the function $f : \mathcal{R} \rightarrow \mathcal{R}$ be defined by $f(x) = x^2 - 3x + 2$ then $f(-3) =$ _____
 (A) 2 (B) 20 (C) -9 (D) 9
6. The value of $\log_2 8 \times \log_2 16 =$ _____
 (A) 12 (B) 4 (C) 128 (D) 7
7. If $\alpha = 60^\circ$, $\beta = 30^\circ$ then $\sin^2 \alpha + \cos^2 \beta =$ _____
 (A) $\frac{3}{2}$ (B) $\frac{3}{4}$ (C) $\frac{1}{2}$ (D) $\frac{1}{4}$

8. Let P be a point on the terminal line of θ ,

then $\sin \theta =$ _____

- (A) $\frac{15}{17}$ (C) $\frac{8}{15}$
(B) $\frac{8}{17}$ (D) $\frac{17}{15}$



9. If $\sin \theta = \frac{3}{5}$ then $\tan \theta =$ _____

- (A) $\frac{4}{5}$ (B) $\frac{4}{3}$ (C) $\frac{3}{4}$ (D) $\frac{5}{3}$

10. $\lim_{x \rightarrow 2} x(x-1)(x+1) =$ _____

- (A) 0 (B) 2 (C) 3 (D) 6

11. The slope of the curve $y = 1 - x^2$ at point $x = 2$ is _____

- (A) 2 (B) -4 (C) -3 (D) 4

12. If $y = x^4 + 12x$ then derivative $\frac{d}{dx}(x^4 + 12x) =$ _____

- (A) $4x^3 + 12$ (B) $x^4 + 12$ (C) $4x^3 + 12x$ (D) $4x^3$

13. If $y = \frac{1}{x}$ then derivative $\frac{d}{dx}\left(\frac{1}{x}\right) =$ _____

- (A) $\frac{1}{x^2}$ (B) $\log x$ (C) $-\frac{1}{x^2}$ (D) $-\frac{1}{x}$

14. The integration of the function $\int 2x dx =$ _____

- (A) x^2 (B) 2 (C) $\frac{x^2}{2} + C$ (D) $x^2 + C$

15. The integration of the function $\int \sin 2x dx =$ _____

- (A) $\frac{\cos 2x}{2} + C$ (B) $\cos 2x + C$ (C) $-\cos 2x + C$ (D) $-\frac{\cos 2x}{2} + C$

16. If $P = \begin{bmatrix} 1 & -2 \\ 2 & 3 \end{bmatrix}$, then $3P =$ _____

- (A) $\begin{bmatrix} 3 & -6 \\ 6 & 9 \end{bmatrix}$ (B) $\begin{bmatrix} 3 & -2 \\ 2 & 9 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & -6 \\ 6 & 3 \end{bmatrix}$ (D) $\begin{bmatrix} 3 & -6 \\ 2 & 3 \end{bmatrix}$

17. If $U = [u_{ij}] = \begin{bmatrix} 2 & -1 & 1 \\ 0 & 1 & 2 \\ 1 & 0 & 1 \end{bmatrix}$ then $u_{23} =$ _____

- (A) -1 (B) 2 (C) 0 (D) 1

18. If $P = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$ and $Q = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$ then $PQ =$ _____
- (A) $\begin{bmatrix} 4 & 2 \\ 9 & 6 \end{bmatrix}$ (B) $\begin{bmatrix} 4 & 3 \\ 6 & 6 \end{bmatrix}$ (C) $\begin{bmatrix} 4 & 2 \\ 6 & 4 \end{bmatrix}$ (D) $\begin{bmatrix} 6 & 9 \\ 3 & 6 \end{bmatrix}$
19. The degree of polynomial $(x^2 - x + 1)(x + 1)$ is _____
- (A) 2 (B) 1 (C) 3 (D) 0
20. The system of equations $x_1 + x_2 = 1$, $x_1 + 2x_2 = 2$ has unique solution using Cramer's rule, if
- (A) $\begin{vmatrix} 1 & -1 \\ 1 & -2 \end{vmatrix} \neq 0$ (B) $\begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} \neq 0$ (C) $\begin{vmatrix} -1 & 1 \\ -2 & 1 \end{vmatrix} \neq 0$ (D) $\begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix} \neq 0$

SECTION II

Q - 2 Attempt any Two of the following.

[10]

- A. Consider the universal set $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ and the sets $E = \{1, 2, 3, 4, 5\}$, $F = \{4, 5, 6, 7\}$, $G = \{5, 6, 7, 8, 9\}$. Find the set relations and construct venn-diagrams of each.

[05]

(a) $E \cup F$ and $E \cap F$

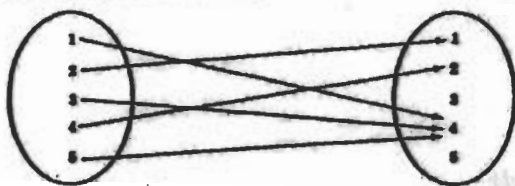
(c) $E \cap F'$

(b) $E \cup G$ and $E \cap G$

(d) $E' \cap F$

- B. Consider the function $f : A \rightarrow B$ defined by the diagram. Find

[05]



(a) $f^{-1}(2)$ (b) $f^{-1}(3)$ (c) $f^{-1}(4)$

(d) $f^{-1}\{1, 2\}$ (e) $f^{-1}\{2, 3, 4\}$

- C. Evaluate the limits (a) $\lim_{x \rightarrow 5} \frac{x-5}{x^2-25}$ (b) $\lim_{x \rightarrow -1} \frac{x^3+4x^2-3}{x^2+5}$

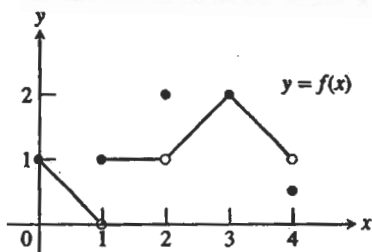
[05]

Q - 3 Attempt any Two of the following.

[10]

- A. Consider the function $y = f(x)$ in figure whose domain is the closed interval $[0, 4]$

[05]



Discuss the continuity of f at $x = 0, 1, 2, 3$, and 4 .

- B. Verify the identity $2 \operatorname{cosec} x = \frac{\sin x}{1 + \cos x} + \frac{1 + \cos x}{\sin x}$

[05]

- C. Simplify the expression $\tan \theta + \frac{\cos \theta}{1 + \sin \theta}$

[05]

SECTION III

Q - 4 Attempt any Four of the following. [20]

A. Find $\frac{dy}{dx}$ if $y = \frac{\cos x}{1 - \sin x}$ [05]

B. If $M = \begin{bmatrix} 3 & 6 \\ 4 & 7 \end{bmatrix}$ and $N = \begin{bmatrix} 8 & -3 \\ 5 & -6 \end{bmatrix}$
Find (a) $|M|$ (b) $|N|$ (c) show that $|MN| = |M||N|$ [05]

C. Find the derivative of $y = \frac{(x-1)(x^2-2x)}{x^4}$ [05]

D. If $P = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 4 \end{bmatrix}$ and $Q = \begin{bmatrix} 1 & -2 \\ -1 & 0 \\ 2 & -1 \end{bmatrix}$ then compute PQ [05]

E. If $P = \begin{bmatrix} 0 & 2 & 0 \\ 1 & 0 & 3 \\ 1 & 1 & 2 \end{bmatrix}$ and $Q = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ then compute
(a) $2P + 3Q$ (b) $3P - 4Q$ [05]

F. Find the first and second derivative of $y = \frac{x^3}{3} + \frac{x^2}{2} + \frac{x}{4}$ [05]

Q - 5 Attempt any Four of the following. [20]

A. Solve the following system of equations
 $x + y + z = 7, x + 2y + 3z = 16, x + 3y + 4z = 22$ using Cramer's rule. [05]

B. Show that $\int (x \cos x + 4x^{-3}) dx = x \sin x + \cos x - 2x^{-2} + C$ by differentiating the right hand side. [05]

C. Evaluate the integral $\int \frac{4 + \sqrt{t}}{t^3} dt$ [05]

D. Evaluate the integral $\int_0^3 (3x^2 + x - 5) dx$ [05]

E. (a) Solve the system of equations $x_1 + x_2 = 1, x_1 + 2x_2 = 2$ using Cramer's rule. [03]

(b) Solve the quadratic equation $x^2 + 5x + 6 = 0$ [02]

F. Evaluate the integral $\int (x^2 + 2x - 3)^2 (x + 1) dx$ by the method of substitution. [05]